

ALERT 4.0 GOVERNMENT - API Documentation

Current State of Microservice API Endpoints

1. Authentication Service

Base URL: `/api/v1/auth`

- `POST /login` - Authenticate a user and receive a JWT token.
- `POST /register` - Register a new user in the system.

2. User & Profile Service

Base URL: `/api/v1/users`

- `GET /user/{userId}` - Retrieve details of a specific user.
- `GET /profile/{username}` - Get the profile for a username.
- `POST /profile` - Update user profile information.

3. Reference Data Service (User Service)

Base URL: `/api/v1/reference`

- `GET /districts` - Retrieve a list of all 38 districts in Tamil Nadu.
- `POST /districts` - Add new district/taluk information (Internal).
- `GET /resources` - Retrieve global resources available.
- `POST /resources` - Add a new global resource.
- `GET /resources/location/{location}` - Retrieve resources for a specific location.

4. Incident Reporting & Escalation Service

Base URL: `/api/v1/incidents`

- `GET /` - Retrieve all incidents.
- `POST /` - Report a new incident (Triggered by VEO). Includes AI processing and Taluk notification.
- `GET /{id}` - Get detailed information about a specific incident.
- `GET /{id}/image` - Fetch the Base64 image payload from MongoDB for the incident.
- `PUT /{id}/status` - Escalate or update the status/level of an incident (Taluk -> District -> Collector).
- `GET /taluk/{taluk}` - Get all incidents reported within a specific Taluk.
- `GET /stats` - Retrieve aggregate statistics of incidents (Total, Active, Resolved, Critical).
- `DELETE /clear-all` - Clear all incidents (Testing/Admin only).

5. AI Prediction Service

Base URL: `/api/v1/ai`

- `POST /predict/{incidentId}` - Send an incident to the Python Llama3 model for severity prediction and advisory extraction.
- `GET /prediction/{incidentId}` - Fetch an existing AI prediction.
- `GET /weather-advisory` - Fetch current active weather advisories generated by the AI.

6. Executive Approval Service

Base URL: `/api/v1/approvals`

- `POST /` - Create a new broadcast approval request (Sent from District EOC).
- `GET /pending` - Retrieve all pending broadcast approvals requiring Collector authorization.
- `GET /collector/{collectorId}` - Retrieve all approvals designated to a specific Collector.
- `PUT /{id}` - Approve or reject the broadcast (Enforces PENDING -> LIVE validation).

7. Broadcast & Alerting Service

Base URL: `/api/v1/alerts`

- `GET /` - Retrieve a log of all executed broadcasts.
- `POST /` - Execute an emergency cell broadcast/radio alert (Must be approved by Collector).

8. Notification Service

Base URL: `/api/v1/notifications`

- `POST /` - Send a push notification/email to an internal user (e.g. Taluk Officer, Collector).
- `PUT /{id}/read` - Mark a notification as read.

9. Analytics & Dashboard Service

Base URL: `/api/v1/analytics`

- `GET /dashboard` - Retrieve aggregated live dashboard state for the State Admin view.
- `POST /snapshot` - Save an analytics snapshot.

Authentication: All endpoints (except `/api/v1/auth/**`) require a valid Bearer JWT token in the Authorization header.

Data Persistence:

- Relational metadata (Users, Incidents, Approvals, Alerts) is stored in **Oracle DB XE**.
- Large binary files (Incident photos) are stored in **MongoDB Atlas** (`alertgov_media` cluster).